Network Modeling and Analysis in Python

Today's goal • 3/3 items

Module 1
Connectivity in Networks

Welcome to the Course

- Welcome to Network Modeling and Analysis in Python
  Video • 5 min
- MADSwPy Certificate Roadmap
  Reading • 10 min
- Course Syllabus
  Reading • 10 min
- Introduction to Jupyter Notebook
  Reading • 10 min
- Meet Other Learners
  Discussion Prompt • 10 min
- **Help Us Learn About You**
  Reading • 10 min

Homophily and Assortativity

- Homophily and Assortativity
  Video • 16 min

Structural Holes

- Structural Holes
  Video • 12 min

K-core Decomposition

- K-core Decomposition
  Video • 8 min
- Knowledge Check: Connectivity in Networks
  Practice Assignment • 35 min

Module 1 Wrap-Up

- Demo: Connectivity NetworkX Tutorial
  Video • 6 min
- Module 1 NetworkX Tutorial
  Lab • 1h
- Module 1 Assignment
  Graded Assignment • 30 min
- Module 1 Programming Assignment: Connectivity in Networks
  Programming Assignment • 3h
- Optional Readings and Resources
  Reading • 10 min

Optional Lesson: Review of Key Concepts

- Review of Key Concept
  Reading • 10 min
- Network Definitions and Vocabulary
  Reading • 10 min
- Network Connectivity
  Reading • 10 min
- Influence Measure and Network Centralization
  Reading • 10 min
- Network Evolution
  Reading • 10 min
- Optional-Review of Key Concepts
  Practice Assignment • 30 min

Module 2
Community Structure in Networks

Module 3
Network Generative Models

Module 4
Models of Diffusion in Networks

Help Us Learn About You

0:14 / 0:23 2x

As part of getting to know you better, we at the University of Michigan have crafted a survey that should only take a few minutes to complete.

Have you already taken the survey as part of completing another course in this series? No need to take this again unless there's something you'd like our team to be aware of.

Take the survey

Thank you in advance for your participation!

Go to next item ✓ Completed

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